

CCNA

COMPLETE CONFIGURATION

PACKET TRACER

By : Caesar

Github : hanezx1

Covers Device Basics, Interfaces, VLANs, Trunking, Inter-VLAN Routing, STP, EtherChannel, Static Routing, OSPFv2/v3, IPv6, NAT/PAT, DHCP, ACLs, HSRP, SSH, Port Security, Device Management, Verification, and Troubleshooting — every command includes syntax, purpose, and a working example.

File Contents

Basic Device Configuration

Interface Configuration

VLAN & Trunking & Inter-VLAN Routing

Spanning Tree Protocol (STP) & EtherChannel

Routing (Static & Dynamic) & IPv6 Routing

NAT (Static NAT & Dynamic NAT & PAT)

DHCP

Access Control Lists (ACL)

HSRP (First Hop Redundancy) & SSH Configuration

Switch Security (Port Security) & Device Management

IPv6 General Commands

Verification Commands

Basic Device Configuration

```
Router>
```

What it does: The default, most restricted CLI mode. Only basic monitoring commands are available.

When to use it: Entered automatically the moment you connect to a device.

Example:

```
Router> enable
Router#
```

■ NOTE

Prompt ends with ">". Type "enable" to move to Privileged EXEC.

Privileged EXEC Mode

```
enable
```

What it does: Unlocks full monitoring commands (show, copy, reload, debug) and access to configuration mode.

When to use it: Used right after User EXEC, whenever you need to see full status or configure the device.

Example:

```
Router> enable
Router#
```

■ NOTE

Prompt ends with "#". Protect this mode with an enable secret password.

Global Configuration Mode

```
configure terminal
```

What it does: Enters the mode used to change device-wide settings such as hostname, passwords, and interfaces.

When to use it: Entered from Privileged EXEC any time a configuration change is needed.

Example:

```
Router# configure terminal
Router(config)#
```

■ NOTE

Shortcut: "conf t". Use "end" or Ctrl+Z to return to Privileged EXEC.

Hostname

```
hostname <name>
```

What it does: Sets the device name shown in the CLI prompt.

When to use it: Configured early on every device so it's easy to identify in a topology.

Example:

```
Router(config)# hostname R1
R1(config)#
```

Enable Secret

```
enable secret <password>
```

What it does: Sets an encrypted password required to enter Privileged EXEC mode.

When to use it: Configured on every device to protect privileged access.

Example:

```
Router(config)# enable secret Cisc0123
```

■ WARNING

Never rely on the older "enable password" command — it stores the password in plain text. If both are set, enable secret always takes priority.

Console Password

```
Router(config)# line console 0
Router(config-line)# password
Cisco123 Router(config-line)# login
```

What it does: Requires a password for anyone connecting through the physical console port.

When to use it: Used to protect local/physical access to the device.

Example:

```
line console 0
password
<password> login
```

VTY (Telnet) Password

```
line vty 0 15
password
<password> login
```

What it does: Requires a password for remote virtual terminal sessions such as Telnet.

When to use it: Required before Telnet access will work on the device.

Example:

```
Router(config)# line vty 0 15
Router(config-line)# password
Cisco123 Router(config-line)# login
```

■ NOTE

"0 15" covers all 16 possible VTY lines. For SSH-only access, see the SSH section.

Service Password Encryption

```
service password-encryption
```

What it does: Encrypts clear-text passwords stored in the configuration using weak type 7 encryption.

When to use it: Used so passwords aren't shown in plain text in "show running-config".

Example:

```
Router(config)# service password-encryption
```

■ WARNING

Type 7 encryption is easily reversible. It stops casual shoulder-surfing, not a real attacker.

Banner MOTD

```
banner motd #<message>#
```

What it does: Displays a message-of-the-day banner before login, often used for legal warnings.

When to use it: Used to warn unauthorized users or identify the device before authentication.

Example:

```
Router(config)# banner motd #Authorized Access Only!#
```

■ NOTE

The delimiter (# here) can be any character, as long as it doesn't appear inside the message.

Save Configuration

```
copy running-config startup-config
```

What it does: Saves the active running configuration into NVRAM so it survives a reboot.

When to use it: Run after every configuration change you want to keep permanently.

Example:

```
Router# copy running-config startup-config
Destination filename [startup-config]? [Enter]
```

■ NOTE

Shortcuts: "copy run start" or the older "write memory" ("wr").

■ WARNING

Forgetting this step means all your changes are lost the next time the device reloads.

Reload

```
reload
```

What it does: Restarts the device, reloading IOS and the startup configuration.

When to use it: Used after major configuration changes or during troubleshooting.

Example:

```
Router# reload
Proceed with reload? [confirm]
```

■ NOTE

Always save your configuration first with "copy run start".

Startup vs. Running Config

```
show running-config
show
startup-config
```

What it does: Running-config is the active configuration in RAM; startup-config is what's saved in NVRAM and loaded at boot.

When to use it: Compare the two to check whether recent changes have actually been saved.

Example:

```
Router# show running-config
```

Copy Commands

```
copy <source> <destination>
```

What it does: Copies files between locations such as running-config, startup-config, flash, or a TFTP server.

When to use it: Used to back up configurations/IOS images or restore them from a server.

Example:

```
Router# copy running-config tftp
Address or name of remote host []? 192.168.1.100
```

Erase Startup Config

```
erase startup-config
```

What it does: Deletes the saved configuration stored in NVRAM.

When to use it: Used to reset a device back to a blank configuration, typically before a "reload".

Example:

```
Router# erase startup-config
Continue? [confirm]
```

■ WARNING

On switches, this does not clear the VLAN database. Also run "delete flash:vlan.dat" for a full reset.

■ Basic Device Configuration Common Mistakes

- Forgetting to save with "copy running-config startup-config" before a reload.
- Setting only "enable password" instead of the encrypted "enable secret".
- Forgetting "login" under a line — without it, a configured password is never actually enforced.

Interface Configuration

```
interface <type> <number>
```

What it does: Enters configuration mode for a specific physical or logical interface.

When to use it: Required before setting an IP address, description, or other interface-level setting.

Example:

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)#
```

FastEthernet

```
interface fastEthernet <slot>/<port>
```

What it does: Refers to a 100 Mbps Ethernet interface, common on older routers and switches.

When to use it: Used whenever the device model has FastEthernet ports.

Example:

```
Switch(config)# interface fastEthernet 0/1
```

GigabitEthernet

```
interface gigabitEthernet <slot>/<port>
```

What it does: Refers to a 1000 Mbps Ethernet interface found on newer routers and switches.

When to use it: Used whenever the device model has Gigabit ports.

Example:

```
Switch(config)# interface gigabitEthernet 0/1
```

Serial Interface

```
interface serial <slot>/<port>/<port>
```

When to use it: Used to connect routers across simulated WAN links in Packet Tracer.

Example:

```
Router(config)# interface serial 0/0/0
```

■ NOTE

The DCE end of a back-to-back serial cable must configure "clock rate".

Loopback Interface

```
interface loopback <number>
```

What it does: Creates a virtual interface that stays up regardless of physical hardware state.

When to use it: Used for OSPF router IDs, testing, or a stable management address.

```
Router(config)# interface loopback 0  
Router(config-if)# ip address 1.1.1.1 255.255.255.255
```

Example:

Interface Description

```
description <text>
```

What it does: Adds a text label to an interface purely for documentation.

When to use it: Used to note what's on the other end of a link, which helps a lot during troubleshooting.

Example:

```
Router(config-if)# description Link to ISP Router
```

Shutdown / No Shutdown

```
shutdown  
no shutdown
```

What it does: Administratively disables ("shutdown") or enables ("no shutdown") an interface.

When to use it: "shutdown" is used on unused ports for security; "no shutdown" is required to bring any interface up.

Example:

```
Router(config-if)# no shutdown
```

■ WARNING

All router interfaces are shutdown by default. Forgetting "no shutdown" is one of the most common Packet Tracer mistakes.

Duplex

```
duplex [auto | full | half]
```

What it does: Controls whether the interface can send and receive at the same time.

When to use it: Normally left on auto; set manually to fix a duplex mismatch with the connected device.

Example:

```
Switch(config-if)# duplex full
```

Speed

```
speed [10 | 100 | 1000 | auto]
```

What it does: Sets the interface's operating speed.

When to use it: Normally left on auto; set manually to troubleshoot a speed mismatch.

Example:

```
Switch(config-if)# speed 100
```

Bandwidth

```
bandwidth <kbps>
```

What it does: Sets the bandwidth value routing protocols use for metric/cost calculations. It does not change the physical link speed.

When to use it: Commonly set on serial interfaces so OSPF calculates cost correctly.

Example:

```
Router(config-if)# bandwidth 64
```

Clock Rate

```
clock rate <speed>
```

What it does: Sets the clocking speed on the DCE end of a serial link.

When to use it: Required on the DCE side of a back-to-back serial connection.

Example:

```
Router(config-if)# clock rate 64000
```

■ NOTE

Check which end is DCE with "show controllers serial 0/0/0".

IP Address (IPv4)

```
ip address <address> <subnet-mask>
```

What it does: Assigns an IPv4 address and subnet mask to an interface.

When to use it: Required on every routed interface that needs to send/receive IPv4 traffic.

Example:

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
```

IPv6 Address

```
ipv6 address <address>/<prefix-length>
```

What it does: Assigns an IPv6 address to an interface.

When to use it: Used whenever IPv6 is enabled on that link.

Example:

```
Router(config-if)# ipv6 address 2001:DB8::1/64
```

■ NOTE

"ipv6 unicast-routing" must also be enabled globally before the router will route IPv6.

■ Interface Configuration Common Mistakes

- Forgetting "no shutdown" — router interfaces are shut down by default.
- Forgetting "clock rate" on the DCE end of a back-to-back serial link, leaving it permanently down.
- Mismatched duplex/speed settings between two connected devices.

VLAN

```
vlan <id>
```

What it does: Creates a new VLAN, or enters configuration mode for an existing one.

When to use it: Used before assigning any ports to a new department or traffic group.

Example:

```
Switch(config)# vlan 10
Switch(config-vlan)#
```

Name VLAN

```
name <vlan-name>
```

What it does: Assigns a descriptive name to the current VLAN.

When to use it: Used right after creating a VLAN, for easier identification later.

Example:

```
Switch(config-vlan)# name SALES
```

Delete VLAN

```
no vlan <id>
```

What it does: Removes a VLAN from the switch's VLAN database.

When to use it: Used to clean up VLANs that are no longer needed.

Example:

```
Switch(config)# no vlan 10
```

■ WARNING

Ports still assigned to a deleted VLAN become inactive until you assign them to a valid VLAN.

Show VLAN

```
show vlan brief
```

What it does: Lists all VLANs on the switch along with the ports assigned to each.

When to use it: Used constantly to verify VLAN creation and port assignments.

Example:

```
Switch# show vlan brief
```

Assign Ports to a VLAN

```
interface <interface>  
    switchport mode access  
    switchport access vlan
```

```
<id>
```

What it does: Puts a port into access mode and places it in a specific VLAN.

When to use it: Used on every port connecting to an end device such as a PC or printer.

Example:

```
Switch(config)# interface fastEthernet 0/1  
Switch(config-if)# switchport mode access  
Switch(config-if)# switchport access vlan  
10
```

Voice VLAN

```
switchport voice vlan <id>
```

What it does: Assigns a separate VLAN for voice traffic on a port that also carries regular data.

When to use it: Used on ports connecting an IP phone with a PC plugged into it.

Example:

```
Switch(config-if)# switchport voice vlan 20
```

Native VLAN (Overview)

```
switchport trunk native vlan <id>
```

What it does: Sets the VLAN whose traffic is sent untagged across a trunk link.

When to use it: Configured to change the native VLAN away from the default (VLAN 1) as a security best practice.

Example:

```
Switch(config-if)# switchport trunk native vlan 99
```

■ NOTE

Full trunk configuration is covered in the Trunking section.

■ VLAN Common Mistakes

- Creating a VLAN but forgetting to assign any ports to it.
 - Assigning a port to a VLAN that doesn't exist yet on that switch.
 - Confusing access ports (single VLAN) with trunk ports (multiple VLANs).
-

Trunking

```
interface <interface>
    switchport mode trunk
```

What it does: Configures a port to carry traffic for multiple VLANs using 802.1Q tagging.

When to use it: Used on switch-to-switch links, or between a switch and a router doing router-on-a-stick.

Example:

```
Switch(config)# interface gigabitEthernet 0/1
Switch(config-if)# switchport mode trunk
```

Allowed VLANs

```
switchport trunk allowed vlan <list>
```

What it does: Restricts which VLANs are permitted to cross a trunk link.

When to use it: Used to limit unnecessary traffic or apply a segmentation policy.

Example:

```
Switch(config-if)# switchport trunk allowed vlan 10,20,30
```

■ NOTE

Use "switchport trunk allowed vlan add <id>" to add a VLAN without replacing the whole list.

Native VLAN on Trunk

```
switchport trunk native vlan <id>
```

What it does: Defines which VLAN is sent untagged across the trunk.

When to use it: Set on both ends of a trunk link to match, usually away from the default VLAN 1.

Example:

```
Switch(config-if)# switchport trunk native vlan 99
```

■ WARNING

A native VLAN mismatch between two switches triggers a CDP warning and can leak traffic between VLANs.

Show Trunk

```
show interfaces trunk
```

What it does: Displays all active trunk ports along with their native and allowed VLANs.

When to use it: Used to verify trunking is configured and operating correctly.

Example:

```
Switch# show interfaces trunk
```

■ Trunking Common Mistakes

- Mismatched native VLANs on the two ends of a trunk link.
 - Forgetting "switchport trunk allowed vlan" and unintentionally allowing every VLAN.
 - Trying to trunk on a port still set to "switchport mode access".
-

Inter-VLAN Routing

```
interface <interface>.<subinterface>
  encapsulation dot1Q <vlan-id>
  ip address <address> <mask>
```

What it does: Creates a logical subinterface that routes for one VLAN over a single trunked physical link.

When to use it: Used when a router (not a Layer 3 switch) needs to route between VLANs.

Example:

```
Router(config)# interface gigabitEthernet 0/0.10
  Router(config-subif)# encapsulation dot1Q 10
  Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```

■ NOTE

The physical interface itself needs only "no shutdown" — it does not get its own IP address.

Multilayer Switch SVI

```
interface vlan <id>
  ip address <address> <mask>
  no shutdown
```

What it does: Creates a virtual Layer 3 interface (SVI) that acts as the gateway for a VLAN.

When to use it: Used on multilayer (Layer 3) switches to route between VLANs without a separate router.

Example:

```
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
```

Enable IP Routing

```
ip routing
```

What it does: Turns on Layer 3 routing capability on a multilayer switch.

When to use it: Required on a multilayer switch before its SVIs can route traffic between VLANs.

Example:

```
Switch(config)# ip routing
```

Default Gateway (Layer 2 Switch)

```
ip default-gateway <address>
```

What it does: Sets the gateway a purely Layer 2 switch uses for its own management traffic.

When to use it: Used on access-layer switches that don't route, so they remain remotely manageable.

Example:

```
Switch(config)# ip default-gateway 192.168.1.1
```

■ Inter-VLAN Routing Common Mistakes

- Forgetting "no shutdown" on the physical interface used for router-on-a-stick subinterfaces.
 - Forgetting "ip routing" on a multilayer switch, so the SVIs never actually route.
 - Mismatched encapsulation VLAN ID between the subinterface and the switch trunk.
-

Spanning Tree Protocol (STP)

```
spanning-tree portfast
```

What it does: Skips the listening and learning STP delays so a port moves straight to forwarding.

When to use it: Used only on access ports connected to end devices like PCs.

Example:

```
Switch(config-if)# spanning-tree portfast
```

■ WARNING

Never enable PortFast on a port connected to another switch — it can create a Layer 2 loop.

```
spanning-tree bpduguard enable
```

What it does: Automatically error-disables a port if it receives a BPDU, protecting against rogue or accidental switches.

When to use it: Used together with PortFast on access ports as a security measure.

Example:

```
Switch(config-if)# spanning-tree bpduguard enable
```

Root Bridge / Priority

```
spanning-tree vlan <id> priority <value>  
spanning-tree vlan <id> root primary
```

What it does: Sets the switch priority to influence, or automatically force, root bridge election for a VLAN.

When to use it: Used to make sure a fast, stable core switch becomes the root bridge.

Example:

```
Switch(config)# spanning-tree vlan 1 root primary
```

■ NOTE

Lower priority wins the election. Default priority is 32768.

Show Spanning Tree

```
show spanning-tree
```

When to use it: Used to verify the STP topology and troubleshoot blocked ports or loops.

Example:

```
Switch# show spanning-tree vlan 1
```

■ STP Common Mistakes

- Enabling PortFast on a port that connects to another switch.
 - Assuming a blocked port is a fault, when it's actually STP correctly preventing a loop.
 - Forgetting that lower priority — not higher — wins the root bridge election.
-

EtherChannel

```
channel-group <number> mode on
```

What it does: Manually bundles interfaces into an EtherChannel without any negotiation protocol.

When to use it: Used when negotiation isn't supported or needed on either end.

Example:

```
Switch(config-if)# channel-group 1 mode on
```

■ WARNING

Both ends must be set to "mode on". A mismatch will prevent the channel from forming.

```
channel-group <number> mode active | passive
```

What it does: Uses the open-standard LACP protocol to negotiate and form an EtherChannel.

When to use it: The preferred method, especially between Cisco and non-Cisco switches.

Example:

```
Switch(config-if)# channel-group 1 mode active
```

■ NOTE

At least one side must be "active" — two "passive" sides will never form a channel.

PAgP EtherChannel

```
channel-group <number> mode desirable | auto
```

What it does: Uses Cisco's proprietary PAgP protocol to negotiate an EtherChannel.

When to use it: Used only between two Cisco devices.

Example:

```
Switch(config-if)# channel-group 1 mode desirable
```

■ NOTE

At least one side must be "desirable" — two "auto" sides will never form a channel.

Show EtherChannel

```
show etherchannel summary
```

When to use it: Used to confirm the channel is up and all expected ports are bundled (status "SU").

Example:

```
Switch# show etherchannel summary
```

■ EtherChannel Common Mistakes

- Mismatched channel modes on each end (e.g., "on" on one side, "active" on the other).
 - Bundling interfaces with different speed/duplex settings — all members must match.
 - Changing port settings after the channel is up instead of configuring the port-channel interface directly.
-

Routing (Static & Default)

```
ip route <destination-network> <subnet-mask> <next-hop-or-exit-interface>
```

What it does: Manually defines a path to a remote network.

When to use it: Used in small networks, stub networks, or for specific paths that shouldn't rely on a dynamic protocol.

Example:

```
Router(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
```

Default Route

```
ip route 0.0.0.0 0.0.0.0 <next-hop-or-exit-interface>
```

What it does: A catch-all route used whenever no more specific route matches the destination.

When to use it: Commonly configured on edge routers pointing toward the ISP or internet.

Example:

```
Router(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

Floating Static Route

```
ip route <network> <mask> <next-hop> <administrative-distance>
```

What it does: A backup static route configured with a higher administrative distance than the primary route.

When to use it: Used to provide automatic failover if the primary static or dynamic route disappears.

Example:

```
Router(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2 200
```

■ NOTE

The AD value (200 here) must be higher than the primary route's AD so it's only used as a backup.

■ Static Routing Common Mistakes

- Typos in the destination network or subnet mask.
- Pointing a static route to the wrong next-hop interface on a multi-access network.
- Forgetting that a floating static route needs a higher AD than the primary route to act only as backup.

OSPFv2

```
router ospf <process-id>
```

What it does: Starts an OSPF routing process on the router.

When to use it: The first step of any OSPF configuration.

Example:

```
Router(config)# router ospf 1
```

■ NOTE

The process ID is only locally significant — it does not need to match between routers.

Router ID

```
router-id <ip-address>
```

What it does: Manually sets a unique identifier for the router within OSPF.

When to use it: Best practice to set manually (often via a loopback) instead of relying on automatic selection.

Example:

```
Router(config-router)# router-id 1.1.1.1
```

■ NOTE

Changes take effect after "clear ip ospf process" or a reload.

Network Statement

```
network <address> <wildcard-mask> area <area-id>
```

What it does: Tells OSPF which interfaces to activate, based on matching the interface's IP address.

When to use it: Required for every network you want OSPF to run on or advertise.

Example:

```
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
```

■ WARNING

This command uses a wildcard mask, not a subnet mask — a very common source of OSPF errors.

Passive Interface

```
passive-interface <interface>
```

What it does: Stops OSPF hello packets from being sent out an interface while still advertising its network.

When to use it: Used on interfaces facing end-user LANs where no OSPF neighbor will ever exist.

Example:

```
Router(config-router)# passive-interface gigabitEthernet 0/1
```

OSPF Cost

```
ip ospf cost <value>
```

What it does: Manually sets the OSPF cost of an interface, overriding the automatic bandwidth-based value.

When to use it: Used to influence which path OSPF prefers when multiple routes exist.

Example:

```
Router(config-if)# ip ospf cost 50
```

OSPF Priority

```
ip ospf priority <value>
```

What it does: Sets a router's priority for DR/BDR election on a multi-access network.

When to use it: Used to control, or completely prevent, a router from becoming the DR.

Example:

```
Router(config-if)# ip ospf priority 0
```

■ NOTE

A priority of 0 means the router can never become DR or BDR.

Default Information Originate

```
default-information originate
```

What it does: Injects a default route into OSPF so other routers in the area learn it.

When to use it: Used on the router that actually has the default/static route to the internet.

Example:

```
Router(config-router)# default-information originate
```

OSPF Show Command

```
show ip ospf neighbor
show ip protocols
show ip ospf interface
```

What it does: Displays OSPF neighbor state, timers, and per-interface OSPF settings.

When to use it: Used constantly to verify adjacencies have formed and troubleshoot OSPF issues.

Example:

```
Router# show ip ospf neighbor
```

■ OSPF Common Mistakes

- Using a subnet mask instead of a wildcard mask in the network statement.
- Mismatched OSPF area IDs between connected routers, which blocks adjacency.
- Forgetting "passive-interface" on LAN-facing interfaces, wasting hello traffic.

IPv6 Routing

```
ipv6 unicast-routing
```

What it does: Enables the router to forward IPv6 traffic between interfaces.

When to use it: Required globally before any IPv6 routing (static or dynamic) will work.

Example:

```
Router(config)# ipv6 unicast-routing
```

Static IPv6 Route

```
ipv6 route <destination-prefix>/<length> <next-hop-or-exit-interface>
```

What it does: Manually defines a path to a remote IPv6 network.

When to use it: Used the same way as an IPv4 static route, just for IPv6 destinations.

Example:

```
Router(config)# ipv6 route 2001:DB8:2::/64 2001:DB8:1::2
```

OSPFv3

```
ipv6 router ospf <process-id>
  router-id <id>
(then on each interface)
ipv6 ospf <process-id> area <area-id>
```

What it does: Configures OSPF for IPv6 networks. Unlike OSPFv2, it's enabled directly on each interface rather than with network statements.

When to use it: Used to dynamically route IPv6 networks.

Example:

```
Router(config)# ipv6 router ospf 1
  Router(config-rtr)# router-id 1.1.1.1
Router(config)# interface gigabitEthernet
0/0 Router(config-if)# ipv6 ospf 1 area 0
```

■ NOTE

OSPFv3 has no "network" command — activation happens per interface.

■ IPv6 Routing Common Mistakes

- Forgetting "ipv6 unicast-routing" globally before configuring any IPv6 routing.
- Assuming a network command works in OSPFv3 the same way it does in OSPFv2 — it doesn't.

NAT

```
ip nat inside
ip nat
outside
```

What it does: Marks an interface as being on the private (inside) or public (outside) side of NAT.

When to use it: Required on every interface that participates in NAT translation.

Example:

```
Router(config-if)# ip nat inside
! (on the LAN-facing interface)
Router(config-if)# ip nat
outside
! (on the internet-facing interface)
```

Static NAT

```
ip nat inside source static <local-ip> <global-ip>
```

What it does: Creates a permanent one-to-one mapping between a private address and a public address.

When to use it: Used for servers that need a consistent, reachable public IP address.

Example:

```
Router(config)# ip nat inside source static 192.168.1.10 203.0.113.10
```

ACL for NAT

```
access-list <number> permit <source> <wildcard>
```

What it does: Defines which private addresses are eligible to be translated.

When to use it: Required before configuring dynamic NAT or PAT. **Example:**

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
```

Dynamic NAT

```
ip nat pool <name> <start-ip> <end-ip> netmask <mask>
ip nat inside source list <acl> pool <name>
```

What it does: Translates a group of private addresses to a pool of public addresses, one-to-one on a first-come basis.

When to use it: Used when you have multiple public addresses to share among internal hosts.

Example:

```
Router(config)# ip nat pool MYPOOL 203.0.113.10 203.0.113.20 netmask 255.255.255.0
Router(config)# ip nat inside source list 1 pool MYPOOL
```

PAT (NAT Overload)

```
ip nat inside source list <acl> interface <interface> overload
```

What it does: Lets many private addresses share a single public address, using different port numbers to tell sessions apart.

When to use it: The most common form of NAT — used whenever only one public IP address is available.

Example:

```
Router(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload
```

Show NAT

```
show ip nat translations
show ip nat statistics
```

What it does: Displays active NAT translations and usage statistics.

When to use it: Used to verify NAT is actually translating traffic as expected.

Example:

```
Router# show ip nat translations
```

■ NAT Common Mistakes

- Forgetting to mark interfaces with "ip nat inside" / "ip nat outside".
- Writing an ACL for NAT that's too broad or too narrow for the intended hosts.
- Confusing PAT (overload) with dynamic NAT — PAT shares one address; dynamic NAT uses a pool.

DHCP

```
ip dhcp pool <name>
```

What it does: Creates a named pool of addresses the router can hand out to DHCP clients.

When to use it: The first step in configuring a router as a DHCP server.

Example:

```
Router(config)# ip dhcp pool LAN-POOL
Router(dhcp-config)#
```

Network for Pool

```
network <network-address> <subnet-mask>
```

What it does: Defines the address range included in the DHCP pool.

When to use it: Required inside every DHCP pool.

Example:

```
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
```

Excluded Address

```
ip dhcp excluded-address <start> <end>
```

What it does: Reserves addresses that DHCP should never hand out, such as those used by the gateway or servers.

When to use it: Configured globally, ideally before creating the pool.

Example:

```
Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

Default Router (Gateway)

```
default-router <address>
```

What it does: Tells DHCP clients which address to use as their default gateway.

When to use it: Required inside a DHCP pool so clients can reach other networks.

Example:

```
Router(dhcp-config)# default-router 192.168.1.1
```

DNS Server

```
dns-server <address>
```

What it does: Tells DHCP clients which DNS server to use.

When to use it: Required for clients to resolve domain names.

Example:

```
Router(dhcp-config)# dns-server 8.8.8.8
```

Lease Time

```
lease <days> [hours] [minutes]
```

What it does: Sets how long a client may keep an assigned address before renewing it.

When to use it: Optional — the Cisco IOS default lease time is 24 hours (1 day).

Example:

```
Router(dhcp-config)# lease 7
```

Show DHCP

```
show ip dhcp binding
show ip dhcp pool
show ip dhcp
conflict
```

What it does: Displays current address leases, pool status, and any address conflicts.

When to use it: Used to verify DHCP is assigning addresses correctly.

Example:

```
Router# show ip dhcp binding
```

DHCP Relay

```
ip helper-address <dhcp-server-address>
```

What it does: Forwards DHCP broadcast requests to a DHCP server located on a different subnet.

When to use it: Used on the client-facing interface when the DHCP server isn't on the local subnet.

Example:

```
Router(config-if)# ip helper-address 192.168.1.5
```

■ DHCP Common Mistakes

- Forgetting to exclude the router's own address, letting DHCP hand it out to a client.
- Forgetting "ip helper-address" when the DHCP server is on a different subnet than the clients.
- Leaving the default-router setting out of the pool, so clients get an address but no gateway.

Access Control Lists (ACL)

```
access-list <1-99> [permit|deny] <source> <wildcard-mask>
```

What it does: Filters traffic based only on the source IP address.

When to use it: Used for simple filtering. Best practice: place close to the destination network.

Example:

```
Router(config)# access-list 10 permit 192.168.1.0 0.0.0.255
```

Extended ACL (Numbered)

```
access-list <100-199> [permit|deny] <protocol> <source> <wildcard> <destination>  
<wildcard> [eq port]
```

What it does: Filters traffic based on source, destination, protocol, and port number. **When to**

use it: Used for precise filtering. Best practice: place close to the source network. **Example:**

```
Router(config)# access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80
```

Named ACL

```
ip access-list [standard|extended] <name>
```

What it does: Creates an ACL identified by name instead of a number, making it easier to read and edit.

When to use it: Preferred over numbered ACLs since individual lines can be added or removed without recreating the whole list.

Example:

```
Router(config)# ip access-list extended BLOCK-TELNET  
Router(config-ext-nacl)# deny tcp any any eq 23  
Router(config-ext-nacl)# permit ip any any
```

Apply ACL to an Interface

```
ip access-group <number-or-name> [in|out]
```

What it does: Activates an ACL on an interface in a specific direction.

When to use it: Required — an ACL has zero effect until it is applied somewhere.

Example:

```
Router(config-if)# ip access-group 10 in
```

■ WARNING

Every ACL ends with an implicit "deny all". Always include at least one permit statement, or all matching traffic will be blocked.

```
access-class <number-or-name> in
```

What it does: Restricts remote management access (Telnet/SSH) based on an ACL.

When to use it: Used to limit which hosts are allowed to manage the device remotely.

Example:

```
Router(config-line)# access-class 10 in
```

Show ACL

```
show access-lists  
show ip interface
```

What it does: Displays configured ACLs, their entries, and how many packets matched each line.

When to use it: Used to verify ACL logic and confirm it's matching the expected traffic.

Example:

```
Router# show access-lists
```

■ ACL Common Mistakes

- Forgetting that every ACL ends with an implicit "deny all".
- Placing a standard ACL too close to the source, blocking more traffic than intended.
- Forgetting to actually apply the ACL to an interface or VTY line with "ip access-group"/"access-class".

HSRP (First Hop Redundancy)

```
standby <group> ip <virtual-ip>
```

What it does: Creates an HSRP group and assigns it a shared virtual IP address used as the default gateway.

When to use it: Configured on two or more routers/multilayer switches to provide gateway redundancy.

Example:

```
Router(config-if)# standby 1 ip 192.168.1.1
```

HSRP Priority

```
standby <group> priority <value>
```

What it does: Sets a router's priority in the HSRP election. Higher priority wins the active role.

When to use it: Used to deliberately control which router becomes active.

Example:

```
Router(config-if)# standby 1 priority 150
```

■ NOTE

Default priority is 100.

HSRP Preempt

```
standby <group> preempt
```

What it does: Allows a router to reclaim the active role once it comes back online with a higher priority.

When to use it: Needed whenever you want the higher-priority router to automatically take back control.

Example:

```
Router(config-if)# standby 1 preempt
```

■ WARNING

Without preempt, a recovered router stays in standby even if its priority is higher than the current active router.

HSRP Version

```
standby version [1|2]
```

When to use it: Used when version 2 features are needed, or to match the peer router's version.

Example:

```
Router(config-if)# standby version 2
```

HSRP Authentication

```
standby <group> authentication <password>
```

What it does: Adds a simple text password so only trusted routers can join the HSRP group.

When to use it: Used for basic protection between HSRP peers.

Example:

```
Router(config-if)# standby 1 authentication cisco123
```

HSRP Tracking

```
standby <group> track <interface> decrement <value>
```

What it does: Automatically lowers a router's priority if a tracked interface goes down.

When to use it: Used so the standby router takes over if the active router loses its uplink/WAN connection.

Example:

```
Router(config-if)# standby 1 track gigabitEthernet 0/1 decrement 60
```

■ HSRP Common Mistakes

- Forgetting "preempt", so a recovered higher-priority router never reclaims the active role.
 - Using the virtual IP address as a real host address elsewhere on the network.
 - Forgetting to match the HSRP version on both routers.
-

SSH Configuration

```
ip domain-name <domain>
```

What it does: Sets the device's domain name, required before RSA keys can be generated.

When to use it: Configured as an early step before enabling SSH.

Example:

```
Router(config)# ip domain-name ccna.local
```

RSA Keys

```
crypto key generate rsa
```

What it does: Generates the RSA key pair used to encrypt SSH sessions.

When to use it: Required to actually activate SSH on the device.

Example:

```
Router(config)# crypto key generate rsa
```

```
How many bits in the modulus [512]:
```

```
1024
```

■ NOTE

A modulus of 1024 bits or higher is recommended.

Local Username

```
username <name> secret <password>
```

What it does: Creates a local user account with an encrypted password for login authentication.

When to use it: Required for SSH, since it authenticates with a username and password rather than one shared line password.

Example:

```
Router(config)# username admin secret Cisco123
```


Local Logic

```
login local
```

What it does: Tells a line to authenticate against local usernames instead of a single line password.

When to use it: Used on VTY lines together with SSH.

Example:

```
Router(config-line)# login local
```

SSH Version

```
ip ssh version 2
```

What it does: Forces the device to use the more secure SSH version 2. **When**

to use it: Best practice, since SSH version 1 has known vulnerabilities.

Example:

```
Router(config)# ip ssh version 2
```

VTY Configuration for SSH

```
line vty 0 15
  transport input
  ssh login local
```

What it does: Restricts the virtual terminal lines to SSH only and enables local username authentication.

When to use it: Used to fully lock down remote management access to SSH-only.

Example:

```
Router(config)# line vty 0 15
Router(config-line)# transport input
ssh Router(config-line)# login local
```

SSH Verification

```
show ip ssh
show ssh
```

What it does: Displays SSH configuration status and any currently active SSH sessions.

```
Router# show ip ssh
```

When to use it: Used to confirm SSH is enabled and working correctly.

■ SSH Common Mistakes

- Trying to generate RSA keys before setting a hostname and domain name (it will fail).
- Forgetting "login local" — without it, SSH prompts for a password but never checks a username.
- Leaving "transport input" as the default (which allows Telnet too), when SSH-only was intended.

Switch Security (Port Security)

Enable Port Security

```
switchport port-security
```

What it does: Turns on port security, which restricts which MAC addresses may use a port.

When to use it: Used only on access ports, never on trunk ports.

Example:

```
Switch(config-if)# switchport port-security
```

■ WARNING

The port must already be in access mode ("switchport mode access") or this command will be rejected.

Maximum MAC Addresses

```
switchport port-security maximum <number>
```

What it does: Sets how many MAC addresses are allowed on the port at the same time.

When to use it: Used to limit a port to a specific number of connected devices, often just 1.

Example:

```
Switch(config-if)# switchport port-security maximum 2
```

Sticky MAC

```
switchport port-security mac-address sticky
```

What it does: Automatically learns and permanently saves the first MAC address(es) seen on the port.

When to use it: Used instead of manually typing every allowed MAC address.

Example:

```
Switch(config-if)# switchport port-security mac-address sticky
```

Violation Mode

```
switchport port-security violation [protect|restrict|shutdown]
```

What it does: Defines what happens when an unauthorized MAC address is detected on the port.

When to use it: "shutdown" (default) disables the port; "restrict" drops traffic and logs it; "protect" drops traffic silently.

Example:

```
Switch(config-if)# switchport port-security violation restrict
```

■ NOTE

A port that's err-disabled after a shutdown violation needs "shutdown" then "no shutdown" to recover.

```
show port-security  
show port-security interface <interface>
```

What it does: Displays port security settings, violation counts, and learned MAC addresses.

When to use it: Used to verify configuration and troubleshoot violations.

Example:

```
Switch# show port-security interface fastEthernet 0/1
```

■ Port Security Common Mistakes

- Enabling port security on a port that's still in dynamic/trunk mode.
 - Forgetting that a shutdown-violation port needs manual "shutdown"/"no shutdown" to recover.
 - Setting maximum MAC addresses to 1 on a port that actually has a phone and PC daisy-chained together.
-

Device Management

```
cdp run
no cdp enable
show cdp neighbors
```

What it does: A Cisco proprietary protocol that shares basic device information with directly connected Cisco devices.

When to use it: Enabled by default; used to discover connected devices, platforms, and their IP addresses.

Example:

```
Router# show cdp neighbors detail
```

■ WARNING

Disable CDP ("no cdp enable" on the interface) facing untrusted or external networks — it can leak device details.

LLDP

```
lldp run
show lldp neighbors
```

What it does: An open-standard alternative to CDP that works between Cisco and non-Cisco devices.

When to use it: Used for neighbor discovery in mixed-vendor environments.

Example:

```
Switch(config)# lldp run
```

NTP

```
ntp server <ip-address>
```

What it does: Synchronizes the device's clock with a time server.

When to use it: Used so log timestamps stay accurate and consistent across all devices.

Example:

```
Router(config)# ntp server 192.168.1.100
```

■ NOTE

Check sync status with "show ntp status" or "show ntp associations".

Syslog

```
logging host <ip-address>
```

What it does: Sends log messages to an external syslog server for centralized storage. **When to use it:** Used in larger networks to keep a permanent, searchable record of events. **Example:**

```
Router(config)# logging host 192.168.1.50
```

Logging (Buffer/Console)

```
logging buffered <size>  
logging console
```

What it does: Controls where log messages are stored or displayed locally on the device.
When to use it: Used to review recent events with "show logging", or to reduce console message clutter.
Example:

```
Router(config)# logging buffered 16384
```

SNMP

```
snmp-server community <string> [ro|rw]
```

What it does: Enables SNMP and sets a community string that acts as a simple password for read-only or read-write access.
When to use it: Used so a network management system (NMS) can monitor the device.
Example:

```
Router(config)# snmp-server community public ro
```

■ WARNING

SNMPv1/v2c community strings are sent in clear text. Use SNMPv3 in real production networks.

■ Device Management Common Mistakes

- Leaving CDP enabled on interfaces facing untrusted or external networks.
- Using SNMPv1/v2c community strings on a production network instead of SNMPv3.
- Forgetting to set NTP, causing log timestamps to be inconsistent across devices.

IPv6 General Commands

```
ipv6 address <prefix>/<length> eui-64
```

What it does: Assigns an IPv6 address where the host portion is generated automatically from the interface's MAC address.

When to use it: Used to quickly assign addresses without typing the full interface ID.

Example:

```
Router(config-if)# ipv6 address 2001:DB8:1::/64 eui-64
```

IPv6 Link-Local Address

```
ipv6 address <link-local-address> link-local
```

What it does: Manually sets the link-local address instead of letting the router auto-generate one.

When to use it: Used for consistent, predictable addressing in labs.

Example:

```
Router(config-if)# ipv6 address FE80::1 link-local
```

Ping IPv6

```
ping <ipv6-address>
```

What it does: Tests reachability to an IPv6 destination.

When to use it: Used to verify IPv6 connectivity, exactly like a regular ping for IPv4.

Example:

```
Router# ping 2001:DB8:2::1
```

Show IPv6 Commands

```
show ipv6 interface brief  
show ipv6 route
```

What it does: Displays IPv6 addressing status per interface and the IPv6 routing table.

When to use it: Used to verify IPv6 configuration and routing.

Example:

```
Router# show ipv6 interface brief
```

Verification Commands

General Device

<code>show version</code>	IOS version, uptime, hardware, and license info.
<code>show running-config</code>	The active configuration currently in RAM.
<code>show startup-config</code>	The saved configuration that loads on boot.
<code>show flash:</code>	Files stored in flash memory, including the IOS image.
<code>show processes cpu</code>	CPU utilization by process.
<code>show clock</code>	The device's current date and time.
<code>show inventory</code>	Installed hardware modules and serial numbers.

Interfaces

<code>show ip interface brief</code>	Quick status/IP summary of all interfaces — the most-used command in CCNA.
<code>show interfaces</code>	Detailed interface statistics: errors, drops, duplex, speed.
<code>show interfaces <interface></code>	Detailed statistics for one specific interface.
<code>show interfaces status</code>	Compact interface status table (switches).
<code>show controllers serial <interface></code>	Shows whether the interface is DCE or DTE.
<code>show ipv6 interface brief</code>	Quick status/IPv6 address summary of all interfaces.

Layer 2 / Switching

<code>show mac address-table</code>	The switch's learned MAC-address-to-port mappings.
<code>show vlan brief</code>	VLANs and the ports assigned to each.
<code>show interfaces trunk</code>	Active trunk ports with native/allowed VLANs.
<code>show spanning-tree</code>	STP root bridge, port roles, and states.
<code>show etherchannel summary</code>	EtherChannel bundle and member port status.

<code>show port-security</code>	Port security settings and violation counts.
<code>show port-security interface <interface></code>	Port security detail for one interface.

Routing

<code>show ip route</code>	The IPv4 routing table.
<code>show ipv6 route</code>	The IPv6 routing table.
<code>show ip protocols</code>	Active routing protocols and their key settings/timers.
<code>show ip ospf neighbor</code>	OSPF neighbor relationships and their state.
<code>show ip ospf interface</code>	Per-interface OSPF settings (cost, priority, timers).
<code>show ip ospf database</code>	The OSPF link-state database (LSDB).
<code>show standby</code>	HSRP group status: active/standby role and virtual IP.

Address Resolution & Neighbors

<code>show arp</code>	The ARP table mapping IP addresses to MAC addresses.
<code>show cdp neighbors</code>	Directly connected Cisco devices (summary).
<code>show cdp neighbors detail</code>	Directly connected Cisco devices, full detail including IP address.
<code>show lldp neighbors</code>	Directly connected devices using the open-standard LLDP protocol.

NAT / DHCP / ACL

<code>show ip nat translations</code>	Active NAT/PAT translation entries.
<code>show ip nat statistics</code>	NAT configuration summary and hit counters.
<code>show ip dhcp binding</code>	Current DHCP leases handed out by the router.
<code>show ip dhcp pool</code>	DHCP pool configuration and usage.
<code>show access-lists</code>	Configured ACLs and per-line match counters.

Security & Management

<code>show ip ssh</code>	SSH configuration status and version.
<code>show users</code>	Currently connected management sessions.
<code>show logging</code>	Buffered log messages stored on the device.
<code>show ntp associations</code>	NTP server sync status.

Troubleshooting Commands

Connectivity Testing

<code>ping <address></code>	Tests basic reachability to a destination using ICMP echo.
<code>ping <address> source <interface></code>	Tests reachability while forcing a specific source interface/address.
<code>tracert <address></code>	Shows the hop-by-hop path packets take to a destination.
<code>telnet <address></code>	Tests TCP reachability/port access to a remote device.

Configuration Review

<code>show running-config</code>	Confirms the currently active configuration.
<code>show startup-config</code>	Confirms what will load after a reload — compare with running-config.
<code>show ip interface brief</code>	Quickly spots interfaces that are down or missing an IP address.

Layer 3 Issues

<code>show ip route</code>	Confirms whether a route to the destination network actually exists.
<code>show arp</code>	Confirms the local device can resolve the next hop's MAC address.
<code>show ip ospf neighbor</code>	Confirms OSPF adjacencies have actually formed.
<code>show ip nat translations</code>	Confirms NAT is translating the expected traffic.
<code>show access-lists</code>	Confirms whether an ACL is unexpectedly blocking traffic (check match counters).
<code>show standby</code>	Confirms which router is active/standby in an HSRP group.

Layer 2 Issues

<code>show mac address-table</code>	Confirms which port a device's MAC address was learned on.

<code>show vlan brief</code>	Confirms a port is in the VLAN you expect.
<code>show interfaces trunk</code>	Confirms a trunk is up and carrying the right VLANs.
<code>show spanning-tree</code>	Confirms whether a port is blocked by STP.
<code>show etherchannel summary</code>	Confirms all expected ports joined the EtherChannel.
<code>show cdp neighbors</code>	Confirms physical/logical connectivity to the next device.
<code>show lldp neighbors</code>	Same as CDP, for non-Cisco neighbors.

Interface Health

<code>show interfaces <interface></code>	Checks for CRC errors, collisions, or a duplex/speed mismatch.
<code>show interfaces status</code>	Quickly spots err-disabled or administratively-down ports.

Debug Commands (CCNA-Safe)

<code>debug ip icmp</code>	Shows ICMP packets as they're processed — useful for ping troubleshooting.
<code>debug ip routing</code>	Shows routing table changes as they happen.
<code>debug ip ospf adj</code>	Shows OSPF adjacency formation in real time.
<code>debug ip nat</code>	Shows NAT translations as they occur.
<code>debug ip dhcp server events</code>	Shows DHCP lease events as they occur on the router.
<code>undebug all (or: no debug all)</code>	Turns off every active debug — always run this when finished.

■ WARNING

Always turn off debugging when finished with "undebug all". Debug commands can generate heavy console output and noticeably slow down a live device — use them briefly and with a clear purpose.

NOTE: